

Exploring the Effects of Self-Avatars on Virtual Reality-Based Reminiscence Therapy for Young Adults

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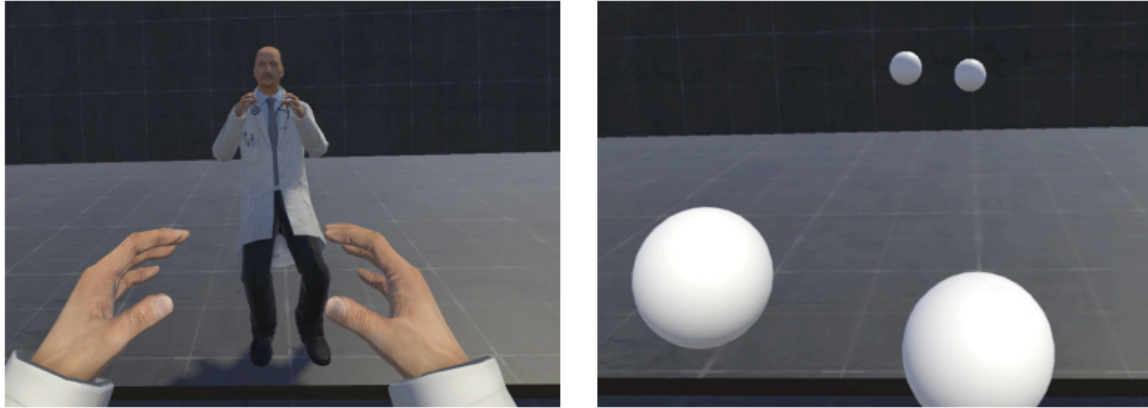


Figure 1: This study investigated the effect of the presence of self-avatar on VR-based reminiscence therapy. In the experiment, young participants recalled events from their lives, with or without self-avatars.

ABSTRACT

Reminiscence therapy is a method to improve mental health by stimulating memories of past experiences and sharing them with others. Virtual Reality (VR) technology has attracted attention as a tool to enhance the effects of reminiscence therapy, as it can provide experiences with a high degree of reproducibility. Meanwhile, VR avatars are known to promote episodic memory, but their effects on reminiscence therapy have not yet been fully investigated. As a preliminary step toward applying VR-based reminiscence therapy for older adults, we conducted a small-scale experiment with eight university students. We found that participants recalled significantly more detailed memories when their VR self-avatars were displayed than when they were not. However, no significant difference was observed in its effect on self-recognition.

Index Terms: Reminiscence Therapy, Virtual Reality, Self-Avatar.

1 INTRODUCTION

Autobiographical memory refers to memories of events and experiences in an individual's life [15]. The recollection of autobiographical memories has been shown to promote re-experiencing associated emotions, enhance extroversion [8] and self-esteem [11], foster language ability [18], and encourage moral and prosocial behavior [9].

One clinical application of these effects is reminiscence therapy. Reminiscence therapy, primarily practiced for elderly people, is a psychological intervention in which individuals or groups

recall past events and share them with others. This therapy has demonstrated benefits such as improved cognitive function [6] and emotional stability [14]. Moreover, reminiscence therapy has been found to be effective for younger individuals as well. For instance, Habermas and Paha (2020) [10] conducted experiments with university students, demonstrating that recalling the past through mementos elicited positive emotions.

Traditionally, reminiscence therapy has utilized cues such as words or photographs. Recently, virtual reality (VR) has gained attention as a potentially more effective method. Tominari et al. (2021) [19] compared reminiscence therapy using VR panoramic images with conventional methods involving still images, demonstrating that VR-based methods not only improved cognitive function as much as traditional methods but also enhanced subjective well-being more significantly. VR provides immersive experiences that enhance the reproducibility of memories, eliciting deeper emotional effects. While previous studies on VR reminiscence therapy have focused on virtual environments, it has also been suggested that virtual bodies, or self-avatars, play a crucial role in episodic memory during VR experiences [5]. Therefore, our study investigates the effects of self-avatars in reminiscence therapy, aiming to further enhance the efficacy of VR reminiscence therapy. This paper reports on a preliminary experiment conducted with university students as an initial step toward applying VR-based reminiscence therapy to older adults.

2 EXPERIMENT

To investigate the influence of self-avatars in VR reminiscence therapy, we conducted an experiment comparing conditions with and without self-avatars while participants performed a task of narrating life events in a VR environment. The experimental design was based on the study by Haj et al. (2020) [7].

The participants were eight university students (seven males and one female, with an average age of 23.9 years). None of the participants had prior knowledge of the experiment, and they were fully informed of the study details beforehand. Upon completion of the

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Figure 2: Self-avatars used in the experiment.

experiment, each participant was compensated with approximately 10 US dollars.

2.1 Experimental Design

The experiment employed a one-factor, two-level within-subject design. Participants experienced both the “Avatar condition,” where a self-avatar was displayed during the reminiscence task, and the “Non-Avatar condition,” where no avatar was displayed. The order of these conditions was counterbalanced among participants.

The self-avatars were selected from the Microsoft Rocketbox Avatar Library¹, widely used in avatar studies. Three avatars with familiar appearances for Japanese participants were chosen: a doctor avatar, a shirted avatar, and a swimsuit avatar (as shown in Figure 2). Each participant used an avatar of the same gender as themselves to eliminate the potential effects of using a different-gender avatar [2]. In the Avatar condition, three different self-avatars were used in random order during the task. In the Non-Avatar condition, self-avatar was not displayed; instead, only a sphere object tracking the participant’s real-world hand movements was shown.

2.2 Procedure

The experiment was conducted in a laboratory at the University of Tokyo. After receiving an explanation of the experiment and providing consent, participants completed a pre-experiment questionnaire, which included three scales: the Nostalgia Scale, Self-Assessment Manikin (SAM), and Psychological Well-Being Scale (PWBS). Participants then donned a Meta Quest 3 headset and performed reminiscence tasks in the VR environment.

The VR environment was created using Unity and designed to be as simple as possible; an inorganic room with gray stone tiles and a virtual mirror for observing the self-avatar. The participant’s head and hands were tracked using the headset and hand-tracking technology, while other body positions and postures were estimated using inverse kinematics calculations with Final IK plugin. To enhance the sense of body ownership and immersion, each recollection task was preceded by 30 seconds spent observing the avatar’s appearance through the virtual mirror and another 30 seconds performing a reaching task.

Each reminiscence task was conducted three times consecutively under either the Avatar or Non-Avatar condition. After a five-minute interval, the remaining condition was conducted three times consecutively. In each task, participants were instructed to “Describe one event from your life in as much detail as possible” within a two-minute time limit. Participants were asked to recall single events lasting less than 24 hours rather than long-term or habitual occurrences and to provide detailed spatial and temporal information.

After each task, participants completed the Igroup Presence Questionnaire (IPQ) to measure VR presence and the Virtual Embodiment Questionnaire (VEQ) to measure body ownership. During intervals and after the experiment, participants completed the

same three pre-experiment questionnaires again, followed by an open-ended interview.

2.3 Measurements and Hypotheses

2.3.1 TEMPau Scale

Specificity of the recalled episodes were scored using the TEMPau scale [12, 3], based on the following criteria:

- 0 points: No memory or only general information is recalled.
- 1 point: A repetitive or long-duration event is recalled.
- 2 points: An event situated in time and/or space is recalled.
- 3 points: A specific event lasting less than 24 hours situated in time and space is recalled.
- 4 points: A richly detailed specific event (including perceptions, emotions, thoughts, and visual imagery) situated in time and space is recalled.

The maximum score for each condition was 12 points (4 points × 3 episodes). Based on the known effects of self-avatars in VR on memory facilitation [5], we proposed the following hypothesis:

H1 TEMPau scores will be higher in the Avatar condition than in the Non-Avatar condition.

2.3.2 Nostalgia Scale

The degree of nostalgia elicited by the reminiscence task was measured using a Japanese version of the Nostalgia Scale [20]. This three-item, six-point scale includes statements such as “1. Right now, I feel very nostalgic,” rated from “1. Not at all” to “6. Very much.” The baseline score was the total score of these items measured before the experiment, and the increase from this baseline after each condition was analyzed. Since self-avatars are expected to facilitate autobiographical memory recall and emotional re-experiencing, we hypothesized the following:

H2 Nostalgia scores will increase more in the Avatar condition than in the Non-Avatar condition.

2.3.3 SAM

Emotions before and after the experiment were measured using the Self-Assessment Manikin (SAM) [4], which evaluates Valence (positive-negative emotional state) and Arousal (emotional intensity). Participants selected illustrations corresponding to their current state. Baseline scores measured before the experiment were used to calculate the corrected scores, representing the changes after each condition. Given the enhanced effect of reminiscence therapy facilitated by self-avatars, we proposed:

H3 Both Valence and Arousal scores will increase more in the Avatar condition than in the Non-Avatar condition.

2.3.4 PWBS

Psychological well-being was measured using the Japanese version of the PWBS [16, 17]. This scale covers six domains—Autonomy, Environmental Mastery, Personal Growth, Positive Relations with Others, Purpose in Life, and Self-Acceptance—with seven items per domain. Responses were scored from 1 to 7 points, and total scores for each domain were calculated. As with SAM, baseline scores were used to calculate corrected scores. Since self-avatars can potentially improve the overall effects of reminiscence therapy, it was hypothesized that:

H4 All six domains of the PWBS will show higher scores in the Avatar condition compared to the Non-Avatar condition.

¹<https://github.com/microsoft/Microsoft-Rocketbox>

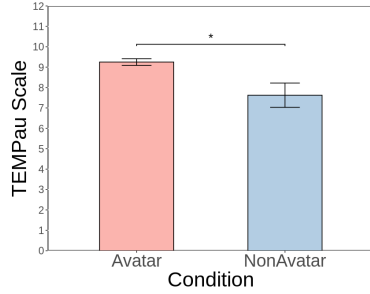


Figure 3: Results of TEMPau Scale (*: $p < .05$)

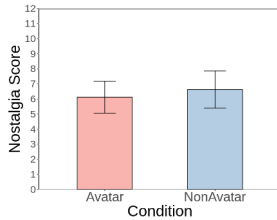


Figure 4: Results of Nostalgia Scale

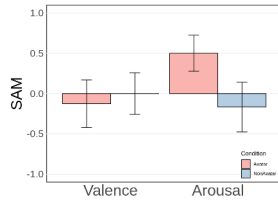


Figure 5: Results of SAM

2.3.5 VR Experience

To assess participants' subjective experiences in VR, we used the IPQ to measure *Presence*, and the VEQ to measure three subscales of the sense of embodiment: *Acceptance*, *Control*, and *Change*. Because displaying a self-avatar in VR has been shown to enhance a sense of immersion [5] and embodiment [13], we propose the following hypotheses:

- H5** Participants in the Avatar condition will report significantly higher *Presence* scores than those in the Non-Avatar condition.
- H6** Participants in the Avatar condition will also report significantly higher VEQ subscale scores (*Acceptance*, *Control*, and *Change*) compared to those in the Non-Avatar condition.

3 RESULTS

3.1 TEMPau Scale

The results of the TEMPau scale scores for the recalled episodes are shown in Figure 3. Since normality was not rejected (Shapiro-Wilk test, $p > 0.1$), a paired t-test was conducted. The specificity scores in the Avatar condition were significantly higher than those in the Non-Avatar condition ($t(7) = 2.39$, $p = 0.048$, $d = 1.32$).

3.2 Nostalgia Scale

The results of the nostalgia scale, measured using a questionnaire, are shown in Figure 4. Since normality was not rejected (Shapiro-Wilk test, $p > 0.1$), a paired t-test was performed. However, no significant differences were observed between conditions ($t(7) = -0.68$, $p = 0.52$, $d = 0.15$).

3.3 SAM

The results of Valence and Arousal scores (both changes from baseline) measured by SAM are shown in Figure 5. The Wilcoxon

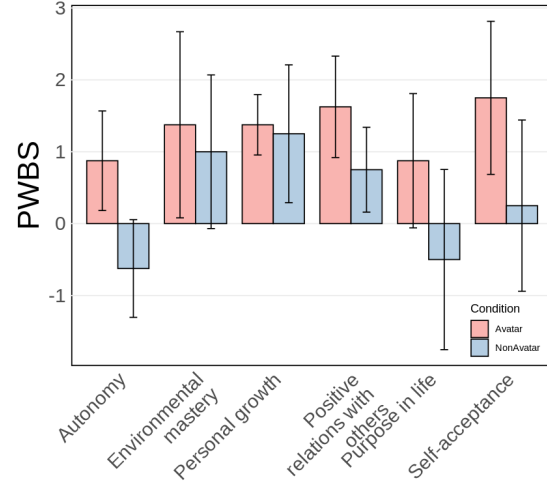


Figure 6: Results of PWBS

signed-rank test revealed no significant differences between conditions for Valence ($p = 1.0$, $r = 0.0$) or Arousal ($p = 0.13$, $r = 0.54$).

3.4 PWBS

The results of PWBS, which measures psychological well-being, are shown in Figure 6. The Wilcoxon signed-rank test was performed for all six domains. No significant differences were observed between conditions for any domain: Autonomy ($p = 0.10$, $r = 0.58$), Environmental mastery ($p = 0.75$, $r = 0.11$), Personal growth ($p = 0.84$, $r = 0.07$), Positive relations with others ($p = 0.41$, $r = 0.29$), Purpose in life ($p = 0.20$, $r = 0.45$), Self-acceptance ($p = 0.31$, $r = 0.36$).

3.5 VR Experience

Wilcoxon signed-rank tests were conducted to evaluate the IPQ and VEQ subscales. Regarding the IPQ value (Presence), participants in the Avatar condition reported significantly higher scores compared to those in the Non-Avatar condition ($p = .01$). Similarly, for the VEQ subscale Acceptance, the Avatar condition also showed significantly higher scores than the Non-Avatar condition ($p < .001$). In contrast, there were no significant differences between the conditions for the VEQ subscales Control ($p = 0.88$) and Change ($p = 0.84$).

4 DISCUSSION

The TEMPau scale scores, which evaluated the specificity of retrieved events, were significantly higher in the Avatar condition. This finding supports H1, which hypothesized that self-avatars would enhance autobiographical memory recall. Additionally, Presence (as measured by the IPQ) was significantly higher in the Avatar condition, supporting H5. This enhanced sense of immersion may have facilitated the more detailed and specific memory recall observed in the Avatar condition, as participants felt more engaged and "present" within the virtual environment [1].

Another possible reason for the increased specificity of retrieved events in the Avatar condition is that self-avatars themselves served as effective triggers, helping participants recall associated memories when they reminisced. In particular, a large proportion of the memories recalled in the Avatar condition were tied to the avatars'

appearances. For example, with the doctor avatar, participants remembered visiting a school nurse or hospital due to injury or illness. With the shirt avatar, they recalled working part-time as a tutor while wearing a shirt, or attending a concert by a singer whose trademark was a shirt. Meanwhile, the swimsuit avatar triggered memories of going to the beach or pool. Interestingly, these memories did not necessarily involve the participants wearing similar outfits themselves; some recalled situations where another person wore a lab coat, which the doctor avatar reminded them of. This outcome may stem from participants perceiving the avatar as an external figure rather than their own body, possibly because they observed it solely through a virtual mirror. Indeed, the VEQ results also reflect this perception: while Acceptance was significantly higher in the Avatar condition, Control and Change did not differ significantly (partially supporting H6). If participants felt they were seeing “someone else” rather than themselves, their sense of agency and transformation would naturally be limited. Strengthening body ownership, e.g., through using more realistic avatar movements, haptic feedback, or personalized avatar designs, may be necessary to observe clearer improvements in these subscales in future studies.

Although the study followed designs used in previous avatar research [13], future studies may need to explore methods that strengthen the sense of embodiment rather than perceiving the avatar as another individual. In contrast to the Avatar condition, participants in the Non-Avatar condition tended to recall more recent or striking life events, likely because of the lack of visual cues. They also seemed to take longer to think of episodes. To clarify these observations, future work could include dependent variables such as the time period of the recalled event and the time participants take before starting to narrate, further illuminating how avatars influence the process of reminiscence therapy.

Regarding H2 (Nostalgia Scale), the current analysis did not reveal a statistically significant difference between conditions. One possible explanation is that participants were young, and the memories they recalled were often quite recent or tied to their student days. This short temporal distance might have made it more difficult for strong feelings of nostalgia to arise. Future studies targeting older adults may be necessary to thoroughly investigate whether self-avatars can facilitate nostalgic experiences in the context of reminiscence therapy.

Although the preliminary experiment lacked sufficient statistical power to detect significant differences, trends were observed where the mean scores for both SAM indicators (Valence and Arousal) and all six domains of PWBS were higher in the Avatar condition than in the Non-Avatar condition. These observations are in line with H3 and H4, which posited that self-avatars would enhance both emotional responses (SAM) and psychological well-being (PWBS). While the differences were not statistically significant in this pilot study, the trends indicate that further research with a larger sample could provide stronger evidence for these hypotheses.

The use of a self-avatar also influenced participants’ subjective experiences in VR. Presence (as measured by the IPQ) was significantly higher in the Avatar condition, indicating that participants felt a stronger sense of “being there” and supporting H5.

One limitation of this experiment is that it primarily involved young adults. Future research will investigate whether the proposed approach also benefits older adults. Accordingly, full-scale experiments and studies targeting older participants are planned to validate the findings. Moreover, several participants’ responses to the questionnaires showed minimal changes across all three measurement points, hinting that reminiscence therapy may have a smaller effect on younger individuals compared to older adults. This might be due to the relatively short time span between the recalled events and the experiment, which in turn leads to fewer changes in self-

awareness. Because of their younger age, many participants naturally focused on events from their student days, which dominated their recollections. Exploring differences in reminiscence therapy outcomes between younger and older individuals remains a novel and potentially important area for future research.

Additionally, certain aspects of the experimental design could be refined. The instructions to recall “life events” may have been overly broad, leading participants to concentrate on either very recent experiences or those from their student days. In fact, some participants reported not knowing what types of memories they were expected to discuss. Adjusting the task content according to participants’ age groups could help elicit a more balanced range of life events. Furthermore, the experimental procedure itself might be improved; previous work often inserted intervals longer than one week between conditions [7, 19], measuring changes in self-state before and after each condition. In this study, by contrast, all three questionnaires (on nostalgia, emotions, and well-being) were administered at three time points (before the experiment, during the interval, and after the experiment). While counterbalancing the order of conditions aimed to mitigate order effects, these effects still seemed influential. A mixed design that treats condition order as an independent variable may therefore offer deeper insights in future research.

5 CONCLUSION

This study aimed to investigate the effects of the presence of self-avatars in VR reminiscence therapy. With a view to examining the influence of self-avatars on VR reminiscence therapy with older adults, a preliminary experiment was first conducted with university students in this study. Participants performed three reminiscence tasks under two conditions: an “Avatar condition,” where self-avatars were displayed, and a “Non-Avatar condition,” where only spheres tracking hand positions were shown.

The hypothesis proposed that the presence of self-avatars would enhance the effects of VR reminiscence therapy and improve the effect of the reminiscence therapy on emotion and psychological well-being, compared to the Non-Avatar condition. The results indicated that self-avatars enhanced the quality and resolution of reminiscence, and while no significant differences were found in self-awareness improvement, there was potential for facilitation.

Based on the findings from this preliminary experiment, improvements in the experimental design have been identified. Future studies will involve larger sample sizes and include experiments targeting older adults to further validate the proposed method.

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